

Assignment 6 Report: Analysis of Trapezoidal Rule Integration Methods

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1 Introduction

This report examines the trapezoidal rule for numerical integration, focusing on three distinct implementations using Python, NumPy, and Cython. Each approach is evaluated based on accuracy, computational efficiency, and memory management. The trapezoidal rule approximates the area under a curve by dividing an interval into smaller subintervals, summing the areas of trapezoids formed by the curve on each segment.

The primary goal is to understand and compare the computational strengths of Python, NumPy, and Cython for numerical integration, assess the performance improvements from vectorization (NumPy) and static typing optimizations (Cython), and evaluate their respective accuracy and efficiency.

2 Implementation Details

2.1 Pure Python Implementation

The Pure Python implementation provides a straightforward iterative approach:

- The integral is computed in a single-pass loop that accumulates trapezoidal areas, avoiding array pre-allocation.
- Minimal function calls are used to reduce execution overhead.

Although easy to implement, this approach suffers from performance limitations due to Python's interpreted nature.

2.2 NumPy Implementation

The NumPy-based implementation leverages vectorized operations for both accuracy and speed:

- `np.linspace` is used for generating equally spaced points.
- `np.trapz` performs the trapezoidal integration using highly optimized array operations, minimizing runtime by handling data arrays in a single step.

This approach provides significant performance gains and an optimal balance between efficiency and code simplicity.

2.3 Cython Implementation

The Cython version introduces C-level optimizations through static typing and loop enhancements:

- `cdef` declarations reduce type-checking overhead.
- Bounds and wraparound checking are disabled, allowing faster looping.
- Direct use of C math functions (e.g., `sin`, `exp`) avoids Python function call overhead.

3 Test Functions

To evaluate the accuracy and performance of each implementation, four test functions of varying complexity were integrated:

1. $f_1(x) = x^2$ (Polynomial)
2. $f_2(x) = \sin(x)$ (Trigonometric)
3. $f_3(x) = e^x$ (Exponential)
4. $f_4(x) = \frac{1}{x}$ (Rational)

4 Results and Analysis

4.1 Accuracy Comparison

Each implementation achieved comparable accuracy, closely matching analytical results. The integration of $f(x) = x^2$ from 0 to 10 with $n = 10^7$ intervals is summarized in Table 1.

Method	Result	Absolute Error
Pure Python	333.333283	5.0×10^{-5}
NumPy	333.333333	$< 1.0 \times 10^{-6}$
Cython	333.333333	$< 1.0 \times 10^{-6}$

4.2 Performance Comparison

Performance was evaluated by measuring execution time for integrating $f(x) = x^2$ with $n = 10^7$ trapezoids, as shown in Table 2.

Implementation	Time (seconds)	Relative Speedup
Pure Python	0.828841	1.0x
NumPy	0.134795	6.1x
Cython	0.366218	2.3x

4.3 Memory and Efficiency Observations

NumPy's vectorization, while efficient in terms of speed, requires larger memory allocations, which may be a consideration for large data sets. The Pure Python and Cython implementations use single-pass accumulation, resulting in reduced memory usage, which is preferable in memory-constrained environments.

5 Challenges

Developing the Cython implementation presented several challenges:

- Module loading issues in Jupyter notebooks were resolved using `%reload_ext Cython`.
- Initially, the Cython implementation used Python's `math` library functions, resulting in performance similar to Pure Python. Switching to C-level functions using `cdef` improved execution speed significantly, although it remains slower than the fully vectorized NumPy approach.
- For compatibility in the final test phase, NumPy functions (e.g., `np.sin`) were required as arguments in `time_trapz`, as non-NumPy functions would result in errors in the `np_trapz` implementation.

6 Conclusion

The choice of implementation depends on the specific requirements for accuracy, speed, and memory:

- **NumPy** provides the most efficient performance due to vectorization and is suitable for high-speed applications.

- **Cython** offers moderate speed improvements over Pure Python, particularly in iterative processing without large array operations.
- **Pure Python**, while least efficient, provides an educational basis for understanding numerical integration.

Future optimizations may involve parallel processing techniques in Cython or using GPU-based computations in Python for even further performance gains.